CHEM 1301 Practice Exam 3-1

Chapters 7, 8, and 9 — 35 multiple-choice questions

Instructions. Choose the single best answer (a–e) for each question. You may use a periodic table. Solubility rules are **not** provided. Calculators are allowed. Answers are rounded to the precision shown. Time yourself: aim to finish in 75 minutes or less.

Answer sheet. Print the bubble answer sheet (`CHEM1301/exam-prep/answer-sheet/answer-sheet.pdf`, landscape, 100% scale) and fill in one circle per question, rows 1–35 (A–E match choices a–e). Leave the rest blank.

1. Which of the following is **NOT** one of the four driving forces that tend to make a chemical reaction occur?

a) Transfer of electrons
b) Formation of water
c) Formation of a solid
d) An increase in the total mass of the products
e) Formation of a gas

2. On the basis of the general solubility rules, which compound is **insoluble** (only slightly soluble) in water?

a) $PbCl_2$
b) Na_2SO_4
c) NH_4Cl
d) $Ba(NO_3)_2$
e) KNO_3

3. Which of the following compounds is expected to be **soluble** in water?

a) Ag_2S
b) $(NH_4)_3PO_4$
c) $Mg(OH)_2$
d) $CaCO_3$
e) $BaSO_4$

4. Aqueous solutions of calcium nitrate, $Ca(NO_3)_2$, and sodium carbonate, Na_2CO_3, are mixed. What is the precipitate?

a) $CaNa_2$
b) No precipitate forms
c) $NaNO_3$
d) $CaCO_3$
e) $Na(NO_3)_2$

5. Which is the correct **net ionic equation** for the reaction of aqueous sodium chloride with aqueous silver nitrate?

a) $Ag^+(aq) + Cl^-(aq) + Na^+(aq) + NO_3{}^-(aq) \rightarrow AgCl(s) + Na^+(aq) + NO_3{}^-(aq)$
b) $Na^+(aq) + NO_3{}^-(aq) \rightarrow NaNO_3(s)$
c) $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + NaNO_3(aq)$
d) $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
e) $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(aq)$

6. When the equation below is balanced with the smallest whole-number coefficients, what is the coefficient of water?

___ $H_3PO_4(aq)$ + ___ $Ca(OH)_2(aq) \rightarrow$ ___ $Ca_3(PO_4)_2(s)$ + ___ $H_2O(l)$

a) 1
b) 2
c) 3
d) 4
e) 6

7. Which pair of aqueous solutions will **NOT** form a precipitate when mixed?

a) KNO_3 and Na_2S
b) $AlCl_3$ and KOH
c) $MgCl_2$ and K_2CO_3
d) $BaCl_2$ and Na_2SO_4
e) $AgNO_3$ and NaCl

8. The net ionic equation for the reaction of **any** strong acid with **any** strong base is:

a) $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$
b) $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$
c) $H^+(aq) + Cl^-(aq) \rightarrow HCl(aq)$
d) $2H^+(aq) + O^{2-}(aq) \rightarrow H_2O(l)$
e) $Na^+(aq) + Cl^-(aq) \rightarrow NaCl(s)$

9. Which of the following is one of the common **strong acids** named in this course?

a) H_3PO_4
b) HF
c) HNO_2
d) H_2SO_4
e) $HC_2H_3O_2$

10. According to the course, which statement best describes a **strong base**?

a) Any compound containing OH, whether or not it dissolves
b) A metal hydroxide that is very soluble in water and completely breaks up into separated ions, so it is a strong electrolyte
c) A metal hydroxide that produces OH^- ions but is not a strong electrolyte
d) A substance that accepts electrons from a metal
e) A substance that produces H^+ ions when dissolved in water

11. In the oxidation–reduction reaction $2Al(s) + N_2(g) \rightarrow 2AlN(s)$, how many electrons are transferred **from each Al atom**?

a) 1
b) 2
c) 3
d) 4
e) 6

12. The reaction $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$ is best classified as:

a) combustion and synthesis
b) decomposition
c) precipitation and double-displacement
d) acid–base only
e) oxidation–reduction and single-replacement

13. Which reaction is a **synthesis** reaction that is also an oxidation–reduction reaction but is **not** a combustion reaction?

a) $2Fe(s) + 3Cl_2(g) \rightarrow 2FeCl_3(s)$
b) $2HgO(s) \rightarrow 2Hg(l) + O_2(g)$
c) $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$
d) $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$
e) $C(s) + O_2(g) \rightarrow CO_2(g)$

14. The mole is defined in this course as:

a) the mass of one atom of carbon, 12.01 g
b) the number of carbon atoms in exactly 12.01 g of carbon, 6.022×10^{23}
c) the number of atoms in 1.000 g of any element
d) the number of molecules in 1 L of any substance
e) 1.66×10^{-24} g

15. Which sample contains the **same number of atoms** as 26.98 g of aluminum?

a) 13.49 g of copper
b) 26.98 g of magnesium
c) 24.31 g of magnesium
d) 6.022 g of hydrogen
e) 12.01 g of magnesium

16. How many moles of aluminum are in 3.60 g of Al?

a) 0.133 mol
b) 1.33 mol
c) 7.49 mol
d) 97.1 mol
e) 8.04×10^{22} mol

17. What is the mass of 2.31×10^{22} atoms of gold?

a) 1.95×10^{-4} g
b) 0.0384 g
c) 7.56 g
d) 5.13×10^{3} g
e) 4.55×10^{24} g

18. How many atoms are present in 4.25 g of magnesium?

a) 1.05×10^{22} atoms
b) 1.05×10^{23} atoms
c) 2.56×10^{24} atoms
d) 3.44×10^{24} atoms
e) 6.22×10^{25} atoms

19. What is the molar mass of aluminum sulfate, $Al_2(SO_4)_3$?

a) 123.0 g/mol
b) 150.0 g/mol
c) 214.2 g/mol
d) 315.2 g/mol
e) 342.2 g/mol

20. What is the mass of 2.45 mol of sodium hydroxide, NaOH?

a) 0.0613 g
b) 16.3 g
c) 56.3 g
d) 98.0 g
e) 143 g

21. How many moles of calcium chloride, $CaCl_2$, are in 9.43 mg of the compound?

a) 8.50×10^{-5} mol
b) 1.25×10^{-4} mol
c) 8.50×10^{-2} mol
d) 1.05 mol
e) 5.12×10^{19} mol

22. What is the mass percent of phosphorus in phosphoric acid, H_3PO_4?

a) 3.09%
b) 12.5%
c) 31.6%
d) 48.4%
e) 65.3%

23. What is the **empirical formula** of glucose, $C_6H_{12}O_6$?

a) $C_3H_6O_3$
b) CHO
c) $C_2H_4O_2$
d) CH_2O
e) $C_6H_{12}O_6$

24. A compound contains 9.94% C, 58.6% Cl, and 31.4% F by mass. What is its empirical formula?

a) CClF
b) $CClF_2$
c) CCl_2F
d) $C_2Cl_3F_3$
e) CCl_2F_2

25. A compound has the empirical formula $C_4H_5N_2O$ and a molar mass of 194.14 g/mol. What is its molecular formula?

a) $C_8H_{10}N_4O_2$
b) $C_{12}H_{15}N_6O_3$
c) $C_{16}H_{20}N_8O_4$
d) $C_4H_5N_2O$
e) $C_8H_{10}N_4O$

26. What experimental information must be known to determine the **molecular** formula of a new compound?

a) Its composition by mass only
b) The mass of the sample only
c) Its composition by mass **and** its molar mass
d) The mass of the sample and its melting point
e) Its molar mass only

27. For the balanced equation $CO(g) + 2H_2(g) \rightarrow CH_3OH(l)$, which statement is correct?

a) 1 mol of CO reacts with 2 mol of H_2 to produce 1 mol of CH_3OH
b) 1 molecule of CO reacts with 2 atoms of H to produce 1 molecule of CH_3OH
c) 3 g of reactants produce 1 g of product
d) 1 g of CO reacts with 2 g of H_2 to produce 1 g of CH_3OH
e) 1 mol of CO reacts with 1 mol of H_2 to produce 1 mol of CH_3OH

28. For $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, which mole ratio converts **moles of H_2** into **moles of NH_3** produced?

a) (2 mol NH_3) / (1 mol N_2)
b) (3 mol H_2) / (2 mol NH_3)
c) (17.03 g NH_3) / (2.016 g H_2)
d) (1 mol NH_3) / (3 mol H_2)
e) (2 mol NH_3) / (3 mol H_2)

29. For $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$, how many moles of Fe can be produced from 17.5 mol of CO?

a) 5.83 mol
b) 11.7 mol
c) 17.5 mol
d) 26.3 mol
e) 35.0 mol

30. The **theoretical yield** of a reaction is:

a) the total mass of all reactants added
b) the amount of product actually obtained in the laboratory
c) (actual yield ÷ theoretical yield) × 100%
d) the amount of product predicted from the amount of the limiting reactant
e) the amount of product predicted from the amount of the excess reactant

31. For $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$, how many grams of SO_3 can be produced from 2.33 mol of O_2?

a) 4.66 g
b) 149 g
c) 187 g
d) 299 g
e) 373 g

32. For $4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$, how many grams of O_2 are needed to produce 54.7 g of NO?

a) 2.28 g
b) 46.7 g
c) 54.7 g
d) 58.3 g
e) 72.9 g

33. For $4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s)$, 17.9 g of Al reacts with 64 g of O_2. Which statement is correct?

a) Al_2O_3 is the limiting reactant
b) Al is the limiting reactant
c) Neither; the reactants are present in exactly the stoichiometric ratio
d) O_2 is the limiting reactant
e) O_2 is the limiting reactant because its coefficient is smaller

34. For $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$, 68 g of Na reacts with 97 g of Cl_2. What is the theoretical yield of NaCl?

a) 2.74 g
b) 80 g
c) 160 g
d) 165 g
e) 173 g

35. For $2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$, 191 g of $KClO_3$ decomposes and 89.9 g of KCl is actually collected. What is the percent yield?

a) 47.1%
b) 51.6%
c) 77.4%
d) 116%
e) 129%